Lean Canvas: AI Traffic Optimization Agent for Smart Cities

SDG 11 - Sustainable Cities and Communities | IBM SkillsBuild AI/ML Internship Final Project

1 PROBLEM

Top 3 problems

- **Congestion wastes time and fuel:** commuters, goods and emergency vehicles lose hours in daily traffic jams.
- **Fixed-timer signals cannot adapt:** they ignore real-time and predicted traffic, so green time is wasted on empty lanes.
- **Pollution and CO_2 rise:** idling vehicles worsen air quality; cities lack affordable, data-driven decision tools.

Existing alternatives: manual policing, fixed-time signals, costly proprietary adaptive systems.

4 SOLUTION

Top 3 features

- **ML forecasting** (LSTM / XGBoost) predicts congestion 15-60 min ahead.
- **Computer vision** (YOLOv8) counts vehicles and lane density from camera feeds.
- **Reinforcement learning agent** (DQN / PPO) optimizes signal timing; A* suggests faster routes; dashboard explains each decision.

8 KEY METRICS

Key activities you measure

- % cut in average waiting time and queue length vs fixed-time signals
- Forecast error (MAE / RMSE); detection accuracy (mAP)
- Estimated fuel and CO_2 saved; travel-time reduction
- Intersections covered / pilot cities

3 UNIQUE VALUE PROPOSITION

Single, clear, compelling message

- **An AI agent that predicts traffic jams before they form and automatically adjusts signals and routes, using existing cameras, at a fraction of the cost of proprietary systems.**
- Less waiting, lower emissions, faster emergency response.

High-level concept: "A self-learning traffic brain for every city."

Early adopter pitch: smart-city pilots and mid-size cities that need results without heavy infrastructure spend.

5 UNFAIR ADVANTAGE

Can't be easily copied or bought

- Prediction + vision + reinforcement learning combined in one agent
- Low-cost, open-source stack that works on existing CCTV
- Models trained on local mixed-traffic data; explainable decisions build trust with authorities

9 CHANNELS

Path to customers

- Pilot projects with municipal bodies and smart-city missions
- Government innovation challenges and tenders
- Open-source GitHub demo, webinars, case studies, LinkedIn
- Partnerships with urban-planning consultants

2 CUSTOMER SEGMENTS

Target customers

- **Primary:** municipal traffic departments and smart-city authorities
- **Secondary:** urban planners and consultants; logistics and fleet operators; ride/navigation platforms
- **Niche pilots:** university campuses, industrial parks, airports, tier-2 and tier-3 cities

End beneficiaries: daily commuters, emergency services and residents of congested areas.

7 COST STRUCTURE

Customer acquisition, distribution, hosting, people

- **Prototype:** near zero (Google Colab, SUMO, Streamlit, GitHub, open datasets).
- **Customer acquisition:** pilot demos, outreach, events and proposal preparation.
- **Hosting and distribution:** cloud GPU/CPU hosting, camera-feed integration, data storage.
- **People:** ML engineers, traffic-domain advisor, support; ongoing model retraining and maintenance.

6 REVENUE STREAMS

Revenue model, lifetime value, revenue, gross margin

- **Revenue model:** SaaS subscription per intersection per month, plus one-time setup and integration fee.
- **Other streams:** analytics dashboard / API licensing, paid pilots and consulting, grants and CSR or smart-city funding.
- **Lifetime value:** multi-year municipal contracts with expansion to more intersections.
- **Gross margin:** high, since it is software-based (target 70-80%, to be validated in pilots).

Lean Canvas format by Ash Maurya. Figures such as margin targets are planning assumptions to be validated through pilots.